

Space as Rendered Boundary: A Geometric Framework for Emergent Spacetime from Event Horizon Structure

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Abstract

This work proposes a geometric framework in which spacetime dimensionality emerges from constrained boundary conditions rather than existing as a fundamental background. Extending prior work in which time was derived from memory topology, we reinterpret time as a one-dimensional event horizon and space as a rendered geometry arising from the interaction between this boundary and an observer-dependent interface.

The framework models the time boundary as a non-orientable manifold with Möbius topology. When constrained by memory structure and rendered through a fiber bundle construction, this boundary geometry generates three-dimensional physical space. The non-orientability induces torsional effects on the constraint manifold that, when rendered into physical space, produce effective gravitational potentials without requiring additional matter fields.

Within this view, phenomena commonly attributed to dark matter may emerge from boundary-induced torsion rather than from undiscovered particles. The speed of light appears as a boundary-limited rendering rate—a maximal constraint on the rendering map rather than propagation through fundamental spacetime. Observer dependence enters through parameterization of the boundary interface, preserving equivalence across observers while avoiding consciousness-based causation.

This proposal offers a coherent structural framework for investigating how spacetime and gravitational effects might arise from constrained boundary geometry. While intentionally speculative, it makes contact with observable consequences through predicted acceleration profiles and provides a pathway toward computational verification.

1 Introduction

In contemporary physics, spacetime is typically treated as a fundamental background upon which physical processes unfold. While remarkably effective, this assumption has motivated growing exploration of whether spacetime itself might be emergent. Approaches ranging from holographic principles and entanglement-based geometry to modified gravity models suggest that spacetime structure may arise from deeper constraints rather than existing as an irreducible arena [2–4].

This perspective becomes particularly salient in the interpretation of dark matter. Despite substantial phenomenological success, the absence of direct particle detection has sustained interest in geometric alternatives—frameworks in which gravitational effects emerge from spacetime structure, information content, or large-scale topology rather than from additional matter constituents [5, 6]. These efforts examine whether foundational assumptions about spacetime and mass require revision rather than denying observational evidence.

A common thread across such perspectives is the increasing importance of boundaries. In quantum mechanics, physical events acquire definiteness through interactions. In relativity, causal structure is defined by horizons and light cones. In gravitational systems, event horizons encode bulk information. These examples suggest that physical structure may be more naturally understood through boundary conditions and constraints than through fully specified interiors.

In prior work [1], the author proposed that time emerges from memory topology rather than existing as a fundamental parameter. Memory was modeled as a constraint structure shaping admissible state transitions, with temporal ordering arising derivatively. This demonstrated that temporal experience can be rendered without reference to an external time coordinate.

The present work extends this perspective by asking whether space itself might be similarly emergent. Specifically, we explore a framework in which time is reinterpreted as a one-dimensional boundary—an event horizon—and experienced space arises as rendered geometry generated from constrained boundary conditions. Apparent spacetime dimensionality results from the interaction between boundary structure and observer-dependent parameterization rather than from multiple fundamental dimensions.

1.1 Scope and Approach

This proposal is intentionally exploratory. It does not seek to replace existing cosmological models, derive a complete theory of gravity, or offer immediate empirical confirmation. Instead, it presents a mathematically coherent structural framework within which space, time, and gravitational effects may be understood as emergent from constrained boundary geometry. The framework makes contact with observable consequences through predicted acceleration profiles, providing a pathway toward computational and observational verification.

1.2 Notation and Conventions

Throughout this work:

- T denotes the time boundary (1-dimensional manifold)
- C denotes the consciousness interface (1-dimensional manifold)
- $\mathcal{M} \subset T \times C$ denotes the memory constraint manifold
- Σ denotes physical space (3-dimensional)
- $F : T \times C \rightarrow \mathbb{R}^3$ denotes the rendering map
- $\mathcal{T}$ denotes the torsion scalar on $\mathcal{M}$
- Φ denotes the effective gravitational potential

2 Boundary Construction and Rendering Bundle

We extend the framework of emergent time derived from memory topology by relocating the topological constraint from internal state space to a boundary geometry. Physical space is not taken as fundamental, but as the rendered image of a constrained boundary manifold.

2.1 Boundary Manifolds

Definition 2.1 (Time Boundary). *Let T denote a one-dimensional boundary manifold associated with time. T is not identified with a flow parameter or coordinate $t \in \mathbb{R}$, but rather with a container-like boundary structure. We model T as a non-orientable manifold with Möbius topology, realized as a circle S^1 with antipodal identification.*

The choice of Möbius topology is motivated by two considerations: (1) it provides the minimal non-orientable one-dimensional structure, and (2) it naturally introduces torsional effects that will prove essential for generating gravitational phenomena.

Definition 2.2 (Consciousness Interface). *Let C denote a one-dimensional manifold representing an observer-dependent selection interface. C parameterizes how boundary configurations are accessed or sampled, without encoding representational content or subjective experience.*

The term "consciousness" is used here in a purely operational sense, denoting the selection or attention coordinate through which boundary configurations become physically realized. It does not imply phenomenal properties, nor does it introduce consciousness-based causation.

2.2 Memory as Boundary Constraint

Definition 2.3 (Memory Constraint Manifold). *Define the memory topology as a constraint manifold*

$$\mathcal{M} \subset T \times C,$$

which specifies admissible boundary configurations. $\mathcal{M}$ encodes structural constraints analogous to those in [1], but now applied to boundary geometry rather than internal state transitions.

This definition generalizes the earlier formulation in which memory constrained internal state evolution. Here, memory constrains which configurations of the (T, C) boundary product are physically realizable. The constraint manifold $\mathcal{M}$ thus plays a role analogous to a gauge-fixing condition or a compatibility constraint in constrained Hamiltonian systems.

2.3 Rendering Bundle

The transition from two-dimensional boundary structure to three-dimensional physical space requires additional geometric structure. We formalize this through a fiber bundle construction.

Definition 2.4 (Rendering Bundle). *Let Σ denote experienced physical space. We define Σ as the total space of a fiber bundle*

$$\pi : \Sigma \rightarrow \mathcal{M},$$

with base $\mathcal{M}$ (2-dimensional) and fiber K (1-dimensional), where K represents render depth or embedding degree. The total space Σ is three-dimensional.

The fiber K may be understood as parameterizing the "depth" at which boundary configurations are rendered into physical space. This could correspond to a radial coordinate, a scale parameter, or an embedding dimension, depending on the specific realization of the framework.

Definition 2.5 (Rendering Map). *Define a rendering map*

$$F : T \times C \rightarrow \Sigma \subset \mathbb{R}^3$$

such that

$$\Sigma = F(\mathcal{M}).$$

The map F is constrained by $\mathcal{M}$ and generates the experienced three-dimensional geometry.

Proposition 2.6 (Dimensional Emergence). *The dimensionality of physical space emerges as:*

$$\dim(\Sigma) = \dim(\mathcal{M}) + \dim(K) = 2 + 1 = 3.$$

This construction avoids the naive dimensional mismatch that would arise from attempting to map a 2-dimensional base directly into 3-dimensional space. The fiber bundle structure provides the additional degree of freedom necessary for the rendering.

3 Non-Orientability and Boundary Torsion

The non-orientable topology of the time boundary induces nontrivial geometric structure on the constraint manifold $\mathcal{M}$.

Lemma 3.1 (Boundary Holonomy). *Parallel transport of vectors along closed loops in $\mathcal{M}$ fails to return vectors to their original orientation due to the non-orientability of T . Specifically, traversing a full cycle in the T direction induces a sign flip in transported vectors.*

Proof sketch. The Möbius identification on T requires that a vector transported around the circle returns with opposite orientation. This holonomy persists under projection to $\mathcal{M} \subset T \times C$, manifesting as nontrivial parallel transport. $\square$

Definition 3.2 (Torsion Scalar). *Let $\mathcal{T}$ denote a torsion scalar associated with the induced connection on $\mathcal{M}$, arising from Möbius identification on the time boundary. Formally, $\mathcal{T}$ measures the failure of local parallelograms to close under the induced connection.*

The torsion scalar $\mathcal{T}$ plays a central role in the framework, as it provides the geometric source term for effective gravitational potentials in rendered space.

4 Effective Gravitational Potential and Dark Matter Interpretation

4.1 Gravitational Potential from Boundary Torsion

Definition 4.1 (Effective Gravitational Potential). *Let Φ be an effective gravitational potential defined on Σ satisfying the field equation*

$$\nabla^2 \Phi = \lambda \mathcal{T},$$

where λ is a coupling constant relating boundary torsion to rendered gravitational effects, and ∇^2 is the Laplacian on Σ .

This equation is structurally analogous to the Poisson equation of Newtonian gravity,

$$\nabla^2 \Phi = 4\pi G\rho,$$

but with the mass density ρ replaced by the geometric torsion scalar $\mathcal{T}$.

Claim 4.2 (Geometric Source of Gravity). *Regions of nonzero $\mathcal{T}$ generate acceleration fields in Σ through the gradient $\nabla\Phi$, without requiring localized particle mass.*

4.2 Dark Matter as Boundary-Induced Torsion

The framework suggests a geometric interpretation of dark matter phenomena:

Claim 4.3 (Dark Matter Interpretation). *Observed gravitational effects attributed to dark matter may arise from regions where the rendering map F induces significant torsion $\mathcal{T}$ on $\mathcal{M}$, rather than from additional particle fields.*

In this view, "dark matter" is not a substance but a signature of boundary geometry. Regions where the constraint manifold $\mathcal{M}$ exhibits nontrivial curvature or where the Möbius twist induces strong holonomy would appear as sources of gravitational acceleration in rendered space.

This interpretation makes contact with observational constraints:

- Galactic rotation curves would reflect the torsional structure of the boundary in regions where $\mathcal{M}$ experiences significant constraint variation.
- Large-scale structure formation would depend on the global topology of $\mathcal{M}$ rather than on particle clustering.
- The apparent dark matter distribution would trace regions of high boundary curvature or torsion rather than mass density.

5 Speed of Light as Boundary Constraint

We interpret the speed of light c as a boundary-limited rendering rate rather than as a velocity through fundamental spacetime.

Proposition 5.1 (Rendering Rate Limit). *If the rendering map F is Lipschitz continuous with constant c ,*

$$\|F(x) - F(y)\| \leq c \|x - y\|,$$

then c represents the maximal rate at which boundary variation may be rendered into experienced geometry.

In this view:

- Light cones correspond to renderability horizons—the boundary of configurations that can be rendered from a given point.
- Null geodesics are optimal paths under the rendering constraint.
- The constancy of c across observers reflects the universal structure of the rendering map rather than properties of a medium.

This interpretation preserves Lorentz invariance at the level of rendered geometry while suggesting a deeper geometric origin for the light speed limit.

6 Observer Dependence and Equivalence

Observer dependence enters the framework through parameterization of the consciousness interface C , but this dependence is formal rather than subjective.

Definition 6.1 (Observer Equivalence Class). *Observers correspond to parameterizations of C . Physically equivalent observers are related by reparameterizations*

$$\varphi : C \rightarrow C$$

that preserve the causal and topological structure of $\mathcal{M}$.

Proposition 6.2 (Observer Invariance). *Physical predictions derived from the rendering map F are invariant under observer reparameterizations φ that preserve $\mathcal{M}$.*

Thus, observer dependence reflects coordinate choice rather than ontological variation. Different observers sample the boundary differently, but the underlying geometry of $\mathcal{M}$ and the structure of the rendering map F remain fixed.

7 Computational Simulation Program

We outline a conceptual pipeline for computational verification:

1. **Discretize boundary coordinates:** Represent $T \times C$ as a discrete lattice or mesh.
2. **Impose Möbius identification:** Implement antipodal identification on the T coordinate.
3. **Construct constraint manifold:** Define $\mathcal{M}$ as a subset of the discretized $T \times C$ satisfying specified constraints.
4. **Define bundle fiber:** Specify the fiber K and the bundle structure $\pi : \Sigma \rightarrow \mathcal{M}$.
5. **Compute induced connection and torsion:** Calculate the torsion scalar $\mathcal{T}$ from the non-orientable structure of $\mathcal{M}$.
6. **Solve field equation:** Numerically solve $\nabla^2 \Phi = \lambda \mathcal{T}$ on the rendered space Σ .
7. **Extract acceleration profiles:** Compute $\mathbf{a} = -\nabla \Phi$ and compare with observed gravitational acceleration curves (e.g., galactic rotation curves).

Even qualitative reproduction of extended gravitational influence—such as flat rotation curves at large radius—would support the plausibility of the framework and suggest directions for refinement.

8 Testable Predictions and Observable Consequences

The framework makes contact with observable phenomena through several channels:

- **Dark matter distribution:** Predicted acceleration profiles should match observed dark matter phenomenology without invoking particle dark matter.
- **Large-scale structure:** The topology of $\mathcal{M}$ should correlate with cosmic web structure.
- **Light speed universality:** The rendering rate limit should remain constant across all observers and reference frames.
- **Compactness hierarchies:** The folding structure of $\mathcal{M}$ should map to observed compactness hierarchies in astrophysical objects.

These predictions are in principle falsifiable through comparison with gravitational lensing data, rotation curve measurements, and large-scale structure surveys.

9 Scope and Limitations

This work intentionally limits its scope:

- It does not replace Λ CDM or claim to resolve all cosmological tensions.
- It does not derive quantum gravity or unify fundamental forces.
- It does not provide immediate empirical validation.
- It does not claim to explain all dark matter phenomenology.

Instead, it proposes a constrained geometric mechanism capable of producing dark-matter-like effects without additional matter fields. The framework is offered as a structural hypothesis intended to support further theoretical and computational exploration.

10 Conclusion

We have proposed a geometric framework in which time is treated as a one-dimensional event horizon and space as a rendered geometry arising from constrained boundary conditions. The non-orientable Möbius topology of the time boundary induces torsional effects on the constraint manifold, which when rendered into physical space generate effective gravitational potentials without requiring additional matter fields.

This framework extends prior work on emergent time from memory topology by relocating the constraint structure to boundary geometry and formalizing the emergence of three-dimensional space through a fiber bundle construction. The resulting picture suggests that spacetime dimensionality, gravitational effects, and even phenomena attributed to dark matter may arise from the geometric structure of boundaries rather than from fundamental fields or particles.

While speculative, the framework makes contact with observable consequences through predicted acceleration profiles and provides a clear pathway toward computational verification. It offers a structural perspective intended to complement existing approaches and to support further investigation into the geometric origins of spacetime and gravity.

References

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